CopperCAM

CopperCAM is a program for Microsoft Windows environment. It provides owners of CNC machines the ability to transform **Gerber** or **Excellon** files, produced by electronic CAD software, to drilling and engraving paths that isolate copper tracks. It can produce a CNC file and directly call up the milling driver with no need to handle intermediate CAM files.

CopperCAM is not an electronic router, not even a printed-circuit CAD system. Its possibilities are not many. It provides no design functions, and consequently focuses on the simple preparatory milling works on isolation engraving and the drilling of a copper circuit that has already been designed.

◘ INSTALLATION

CopperCAM is available only as download from www.coppercam.com website and is not provided on a physical support. The downloaded module is an auto-extractible installation program that allows you to choose the hard disk directory where the software will be installed. Installing makes **no changes in the registry**, except for creating shortcuts on the desktop and Windows "Start" menu.

Note: the downloadable block is programmed for unzipping in directory "**C:\ZGTEMP**" of hard disk **C:** and you cannot use an alternate one. If this directory cannot be created because access is denied for some reason, then installing is cancelled and an error message pops up. If you cannot allow the installer create that directory, then the simplest is to install on another PC, copy the installation and manually create the shortcut icon on the desktop.

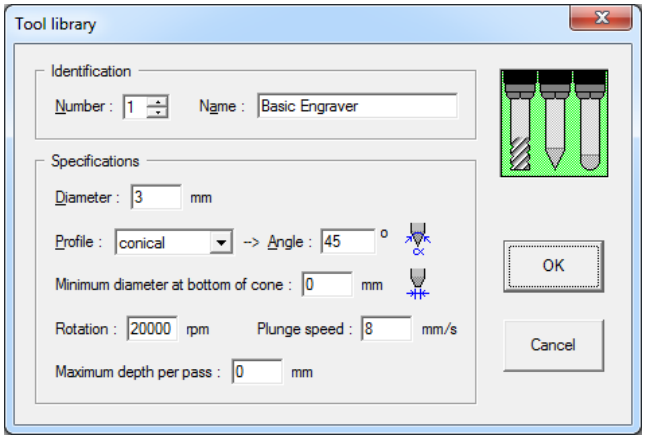
Other note: the directory you have chosen for installing the software must be accessible to the user for write operations, since the application workspace will be saved in it. Do not install in **"C:Program Files"** if the user will be granted limited rights not including write permission. The simplest is to install into **"C:\CopperCAM"** as suggested by default.

◘ LICENCE

CopperCAM licence is defined by a 20 digit code that corresponds to user's personal data (name and address). This licence is sent as a file named **COPPERCAM.LIC** that is to be copied directly into the installation directory, default location being **"C:\CopperCAM"**. A start-up, the program will detect the presence of the licence file and will directly transfer the information to the licence manager. Calling "Help / Licence" lets you check the validity of your licence.

❑ CUTTER TOOLS

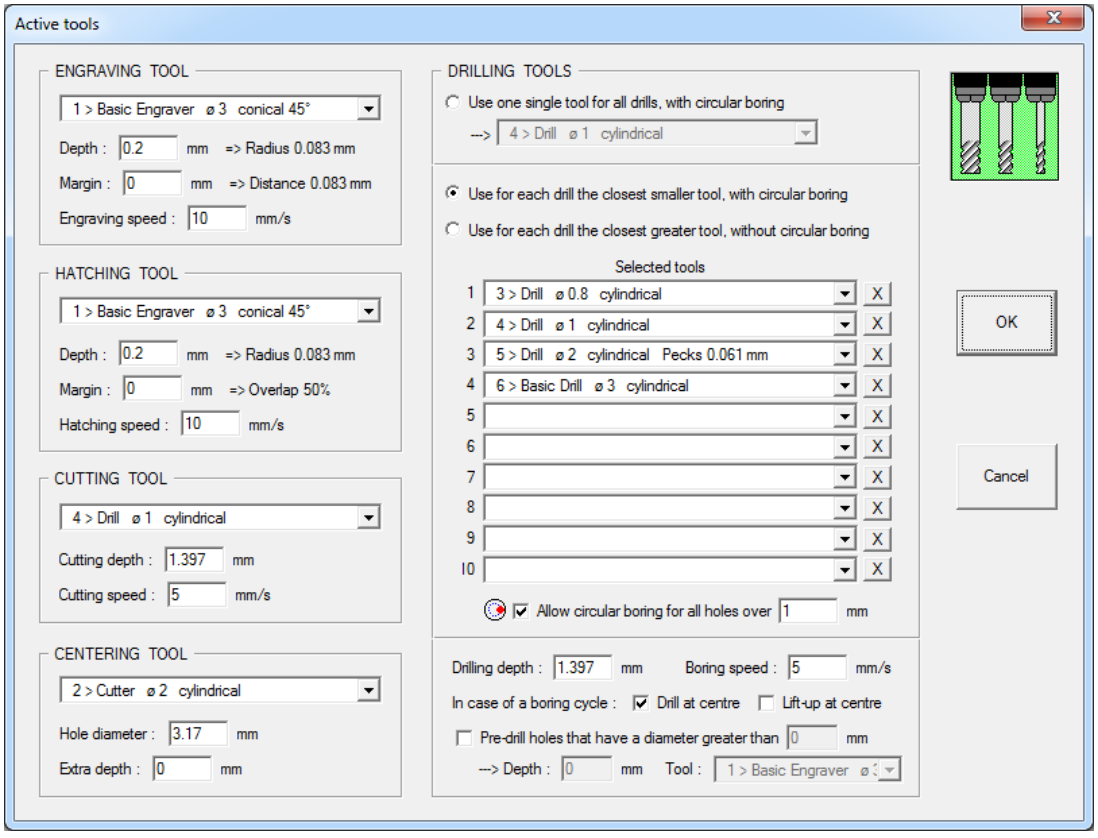
The very first thing to think about once CopperCAM is installed is **define the accurate characteristics of the tools** that will be used for drilling and milling printed-circuit boards. Function "Parameters / Tool library" lets you indicate useful data for each tool. If you have a tool rack, number them and use the same numbers in the parameters, though each tool can be assigned a name.



A tool is defined only if it has been given a diameter and a profile. For a conical engraver, the diameter should be measured at top of the cone (maximum diameter). The cone angle should be measured **from edge to edge** and not from edge to vertical axis. The minimum diameter at bottom of the cone is for a possible flat end.

The maximum depth per pass determines the **milling pecks**: if the Z coordinate is greater, then CopperCAM will automatically manage successive passes at increasing depth until the target Z is reached.

Once tools are defined, it is necessary to **indicate which ones will be assigned different tasks** the program can offer: isolation engraving, drilling, hatching zones, card contouring, and centring on plots. The function "Parameters / Selected tools" gives access to a setup window where, for each task, you must indicate which tool is to be called, and its motion depth and speed. For isolation engravings, it is possible to add a small distance margin.



For isolation engravings, it is possible to setup a small margin for adjusting the apparent tool radius at depth without changing its data in the library. For hatchings, **the margin gives the mesh density**,

i.e. the recovery between two hatch lines. Default value is 50 %, so each new hatch line recovers half width of its neighbour line.

Keep minded that passes for engraving, hatching and cutting are defined by the maximum depth per pass that has been indicated for each tool in the library.

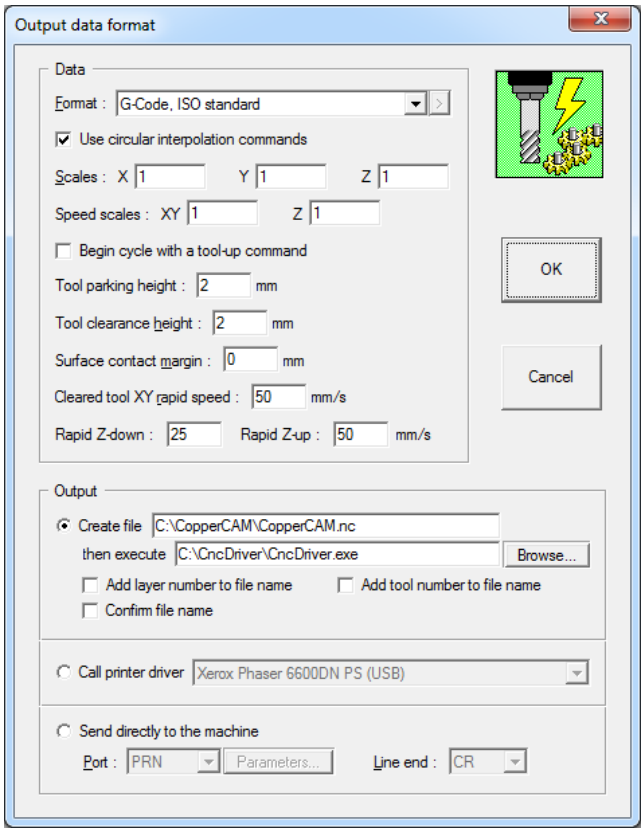
It is possible to **limit the number of drilling tools** and consequently tool change operations, by driving boring cycles when the hole diameter is greater than the tool diameter. However, this function should be used carefully: most drilling bits for epoxy are indeed not qualified for horizontal milling operations, so there is an important risk of break.

Three strategies can be followed for drilling: either use **one single tool** for all drills, with boring cycles if the hole is greater than the tool; or use **a limited number of selected tools** with, for each hole, a call to the tool which has the closest diameter, smaller (with circular boring to reach the diameter) or greater (so without boring).

Drills with big diameters can be preceded by a pre-drill hole at small depth for centering, with the same drill bit or another one more specialised in pointing. In this case, one more tool change will be called by the software.

❑ OUTPUT

Function "Parameters / Output data format" allows you to choose the type of the data which will be produced at last for chaining to the milling driver. Different usual formats are offered, including a **fully definable post-processor** format (see below).



All path coordinates are always relative to the workpiece origin point, depending on the operating mode of your milling driver. There are no machine zero points known to CopperCAM and therefore no absolute coordinates. We will see below how XY and Z origin points should be set.

If the option "Begin cycle with a tool-up command" is enabled, then the very first motion command will send the tool to the Z clearance position above the workpiece. If that option is disabled, then the first motion command will actually move the tool horizontally to XY target position above the path entry point, which supposes that your machine always starts a cycle with the tool cleared high enough for avoiding collisions.

The **surface contact margin** stops the tool rapid plunge before it touches the workpiece surface. So the tool remains a little bit above for beginning its drilling plunge.

Rapid speed for XY cleared motion, Z lowering and lift-up speeds will possibly be ignored by your machine driver, depending on the output format it uses. For example, in G-code, which is the most

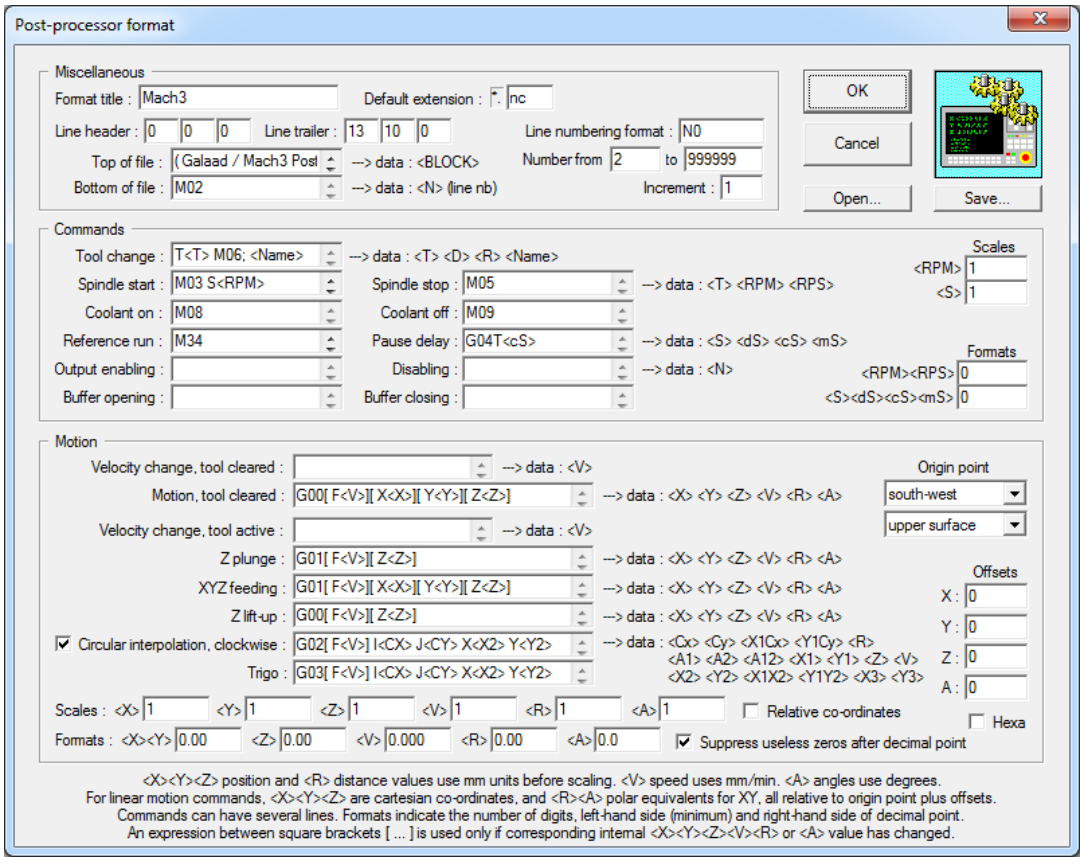
widely used, all G0 motion speeds are supposed to be defined by the operator on the machine console, and not set by the file. In this case, setting these rapid speeds will have no effect.

CopperCAM can **create the output file and then immediately call the driver or transmission software**. This file is not of great interest since it is only an intermediate between CopperCAM and the machine driver. It will be overwritten at every new output, except if no name has been indicated. In such case, you will be prompted to give a name every time. Note that the file name is added automatically as argument to the executable program that is called. So it is useless to add it in its command line.

It is also possible to automatically add the layer number or the tool number to the name of every file created. Then, instead of creating one single file per cycle, CopperCAM will create as many different files as there are layers or tools being used, with identical basic names and different suffixes. The option "Confirm file name" will prompt the operator to validate or modify the name for every new output file.

Some CNC engraving machines are provided with a driver which displays them as **virtual printers** for Windows. In such cases, CopperCAM can also call this printer driver and pass the produced data onto it. It is also possible to output the produced data on a COM or LPT port for a direct link to the machine. This is one-direction only; CopperCAM will not read any feedback.

Now let us come back to the **post-processor**, last line of the available formats. When you select it or click on the small button **>** at top-right, a sub-window pops up for creating or modifying the format. That window is indeed a bit loaded:



We are not going to plunge deep into details. What you should understand first is that each command line can contain variables with their identifiers framed by **< >** signs. These variables will be replaced by corresponding values when producing the output file. Everything that is not variable will be used as it is for building the command line. Furthermore, expressions between square

brackets **[ ]** and containing a variable will be used only if the value of the variable inside has changed. Example for a G-code line for an active feeding motion:

G1[F<V>][X<X>][Y<Y>][Z<Z>]

G1 header will be used anyway. Next expression **[F<V>]** will be added behind only if the value of the variable **<V>** (feed speed) has changed, compared to the last line that integrated it. Then, each expression **[X<X>][Y<Y>][Z<Z>]** will also be added to the line if the variables they contain, respectively **<X> <Y> <Z>**, have its value changed. For example, if the feed speed is now 600 mm/mn, X coordinate is 98.76, Y coordinate is 54.32 and Z coordinate is unchanged, then the line finally assembled by CopperCAM will be **G1 F600 X98.76 Y54.32**

Consequently, **the settable post-processor allows you to define any format** that is based on text lines with decimal or hexadecimal data. Coordinates are submitted to individual scaling factors and the number of decimal digits is also settable: for example, format indicator 000.00 means that we want at least 3 digits at left before the decimal point, and maximum 2 digits at right after the decimal point. The software will manage the rounding. Speaking about this, the decimal point is not the one Windows regional setting provides, but the character you set in this format. For most data formats, including G-code, the decimal separator is the point.

For one single command, **you may enter several lines**. Globally, each line generated by CopperCAM can begin and end by a maximum of three ASCII characters that are to be indicated here at top of the window in decimal code (for example 13 and 10 for CR and LF as line trailer, code 0 being ignored). Every line can be numbered with a formatted header.

You may save your post-processors. A few canvas are available with the software.

◘ GENERAL WORK SEQUENCE

The process consists in **opening a Gerber file** that has been produced by an electronic CAD software, then eventually one or more additional layers (including the card cutting contour) and an Excellon file that contains all drills. CopperCAM can automatically detect a track net that corresponds to the card final cut-out. **It is better to have the card contouring path in a circuit layer** instead of one separate file which may be difficult to realign whenever coordinates appear shifted. Aligning requires pads in the layer.

Tracks that actually correspond to texts, logos and others may be **engraved as a simple centerline**, instead of making a complete isolation of its borderlines. The simplest way to do this is **click on one of them with the right mouse button** and, in the context menu, call function "Engrave tracks as centerlines".

Once layers are imported and aligned, the software will **calculate isolation contours** around pads and tracks (function "Machining / Contours / Calculate contours" or its corresponding shortcut icon). The operator can then rectify the result if necessary, by deleting or adding isolation segments and adding hached zones so all the copper is removed there. Once calculating and rectifying jobs are achieved, the operator calls function "Machining / Mill" (or its shortcut icon). The software then produces an output file as defined in parameters and eventually chains to the module that will supervise the dialog with the machine.

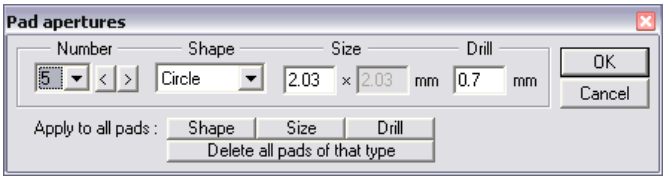
❑ GERBER FILES

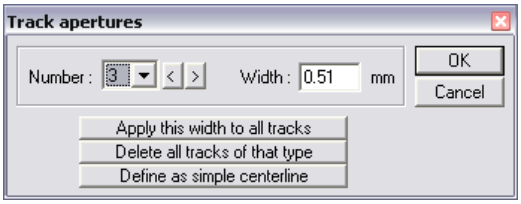
The most widely known in electronic CAD is Gerber format. Designed for Gerber Scientific Instruments photo-plotters, it has become a standard in the area. The photoengraving of circuits using light-printing techniques induces specific considerations that appear in the format. Light-printing is performed by an optical head that focuses the light beam on the circuit after travelling through a diaphragm, at the locations where the copper should be preserved. The diaphragms generally have a predefined size and shape, the simplest being a basic circular disc. The pads may have more exotic shapes, but the tracks are printed using simple circular diaphragms of given diameters along the connecting path.

So a Gerber RS274-D file (old format) contains **diaphragm numbers** that are either predefined in the optical library of the photo-plotter, or referenced in the file itself or in an attached library file. Light-printing instructions are very simple: the flash head can be **moved, with the diaphragm shut**, to a given XY position (movement without light-printing), or **with the diaphragm open** (light-printed track), or even be sent to a position and the **diaphragm being opened then shut** to light-print a fixed point (pad). This makes a total of three positioning instructions, plus the number of the diaphragm that is currently used.

A more elaborate format, which keeps ascending compatibility with the one mentioned above, has been defined under the name Gerber RS274-X or Extended-Gerber. This format keeps the same light-printing instructions, but its major advantage is that it integrates in the file header all **geometrical indications about used diaphragms and even sometimes about drills**. In fact, a Gerber RS274-X file requires neither an attached library file nor a table of predefined diaphragms. Any useful data that is required to create the printed circuit is contained in the file. Naturally, CopperCAM can read this heading information whenever available. If your own electronic CAD application offers an export function under Gerber RS274-X format, it is the one you should use.

Should you open a classical Gerber RS274-D file, no information related to diaphragms is available, and consequently you must redefine the geo-metrical properties of these diaphragms once the file is loaded, *i.e.* for each set of pads indicate the **shape** and the **size** of the diaphragm, and even the **drill** diameter if necessary, and indicate for each set of tracks, the **size** of the diaphragm, which corresponds to the track **width**. This must be done for each diaphragm referenced in the open file.





CopperCAM memorises your indications for the next file, so it is not necessary to redefine pad and track diaphragms at anytime if your electronic CAD application always reuses the same references. Obviously, this task has no object if you are using the Gerber RS274-X format, whose heading indications are picked up and immediately applied. However, as soon as the file is loaded, if a diaphragm that is used in the file is not referenced, then CopperCAM displays a dialogue box which allows you to complete the missing references.

So, in an extended Gerber file, it is possible to get not only the geometrical design information about pads and tracks, but also the drilling diameters of pads for soldering traditional components (component side and colder side). The only information that is still missing – and this is bad news for owners of a CNC machine – is the contour of the whole printed circuit board, with its origin points. Card cut-out path is generally a continuous line of tracks that globally contain the circuit.

Important: the **ground planes** or copper zones can be either made of a thin hatching of classical tracks, or a bounded surface which borderlines are actually tracks using tags G36 and G37 of Gerber RS274-X format. Isolation of thin hatch lines one by one may require hours of calculation and then CopperCAM looks blocked. Of course it is much better to setup the CAD software ahead so it produces polygon surfaces G36 which will require only a few milliseconds for calculating isolation contours. In your CAD application, that option is generally available among Gerber output preferences.

❑ EXCELLON FILES

Another file format concerns **pad drilling** in a printed circuit, this format has been designed for Excellon Automation multi-drills. Like Gerber files, Excellon files contain XY drilling coordinates and tool numbers whose diameters correspond to holes. And like old GRB files, EXL (or sometimes DRL or TXT) files unfortunately do not always contain this drilling diameter information, so it is necessary to fetch it in attached files that are not standard, or indicate it manually once the file as been loaded.

Furthermore, there are actually two different Excellon formats, one using real co-ordinates (XY numerical values are formatted and indicated in immediately usable units), the other being a bit older in co-ordinates with no trailing zeros on the right-hand side, which may induce position errors if the file does not contain header information about the numeric framing format. To avoid this problem, CopperCAM lets you select both formats under two distinct entries, which allows you to directly choose the correct numerical model. If your file looks wrong, load it again using the other available Excellon format, and all being well things should look better.

An Excellon file cannot represent a printed circuit, not even its pads, but only its drills. Note that the Gerber RS274-X format may encode drilling diameters for pads, but this is optional and unfortunately not often used by CAD software. It is better to include drilling data to the Gerber file if your software can. This avoids handling a supplementary Excellon file and eventually realign drills with pads.

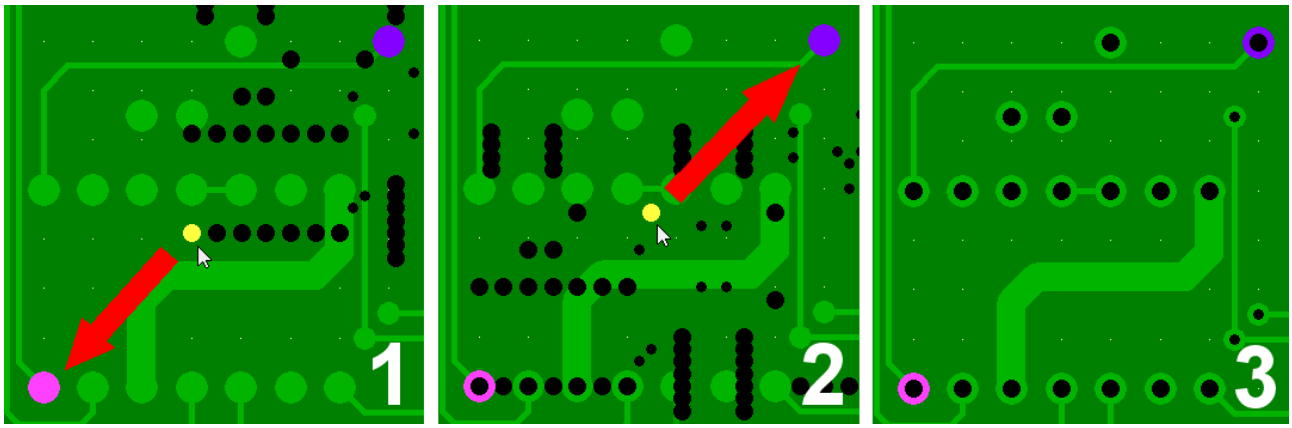
❑ ALIGNING AND ADJUSTING LAYERS

CopperCAM is able to manage **four copper layers** simultaneously (pads and tracks), plus a **drilling plane** (layer #5) and a **card contour plane** (layer #6). Gerber or Excellon, open files do not always use the same origin point. Consequently, it may become necessary to **realign items** so they match one another. The first thing to do is rotate the current layer so it has the same orientation as the layer #1, by using the inverting and rotating commands.

On layer #1, select a **reference pad**. Then skip to another layer (circuit or drills) and select a pad or a drill to be aligned to the reference. That can be done directly with the **right mouse button** on a pad, or with command "File / Reference pas #1 / Define" or its shortcut icon on top bar.

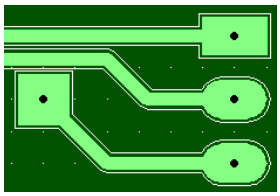
Reference pad #1 can be used only for realigning layers with a simple offset, considering that they already have the correct scale and orientation. But it is also possible to rotate and rescale a layer which does not match the reference layer, for example the drill layer which often comes from a file under a different format with incompatible orientations and dimensions. The procedure is simple: in the reference layer (the one which is supposed to have the correct scale), just plot two reference pads, if possible a bit distant from one another. Then skip to the layer to be realigned (circuit or drills) and first adjust the pad or drill on the **reference pad #1 (pink)** which corresponds, so the layer is globally repositioned according to that reference pad. Now adjust the other pad or drill on

the **reference pad #2 (violet)** which corresponds, to globally rotate and rescale that layer. Alignment and scaling are then made easily, with no need to reset the import scale.



Do not confuse **origin** and **reference pad**. The origin is the point of co-ordinates (0,0) of the circuit and consequently of the output milling file. The reference pads are used only for realigning different layers. They do not affect coordinates.

❑ CALCULATING ISOLATION CONTOURS

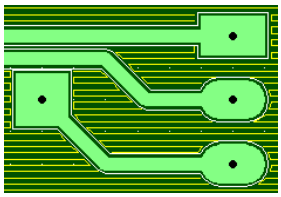


Isolating tracks and pads requires a preliminary calculation that defines **path contours** according to the end diameter of the cutter tool that will be used for the engraving task. This calculation manages collisions should trajectories overlap. This is the main purpose of CopperCAM which, from the geometrical information the Gerber file contains, will be able to create isolation paths and even allow you to remove the copper from all empty zones using hatchwork. These functions can be applied from "Machining / Contours / Calculate contours" (or the corresponding shortcut icon).

The distance between the tool axis path and the track or pad border is generally the engraving **tool end radius**. CopperCAM then calculates all isolation contours for engraving the circuit. This calculation is performed in several phases that may sometimes take some time depending on the complexity of the circuit and the computer speed.

Note: it is possible to correct an isolation path after it has been calculated, using other sub-functions in "Machining / Contours" (or corresponding icons). You can **add an isolation segment** that could not be inserted between two pads or tracks that were too close, or **delete all or part of a useless isolation path**, for example to connect a pad or a track net to a ground plane. The simplest for learning these is try and error, keeping in mind that any operation can be undone.

❑ HATCHING EMPTY ZONES



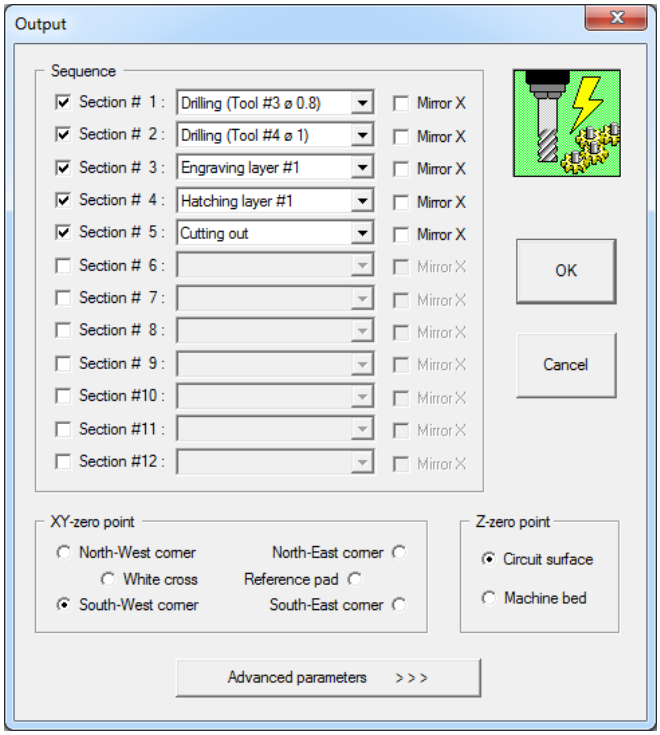
Engraving contours is sufficient to isolate tracks and pads, but you may want to extend the job and **remove the remaining copper** from the card surface or part of it, to ease soldering components, for example on very neighbour pads.

The hatching function can be called up only if the **track and pad contours have already been calculated**. The hatch margin is defined in selected tool parameters. It corresponds to the distance between two consecutive lines. The default value is the tool radius at the engraving depth (for a conical tool), which gives a recovery ratio of 50%. But you may of course choose a different margin. CopperCAM can also link hatch lines to build a zig-zag path whenever possible without milling a shortcut through a contour. Hatch ends approach the contours at about 10% of the hatching distance.

Hatching is cartesian, using horizontal or vertical direction. If option "Optimal" has been selected, then the software calculates both directions and keeps the one that gives the shortest cycle. So it is longer to calculate but faster to mill.

❑ MILLING

Calling function "Machining / Mill" produces the output file and automatically chains to the machine driver if it has been set as an executable program. The job sequence is definable through different sections, each of them corresponding to a task (drilling, engraving different layers, drilling centring plots, final cut).

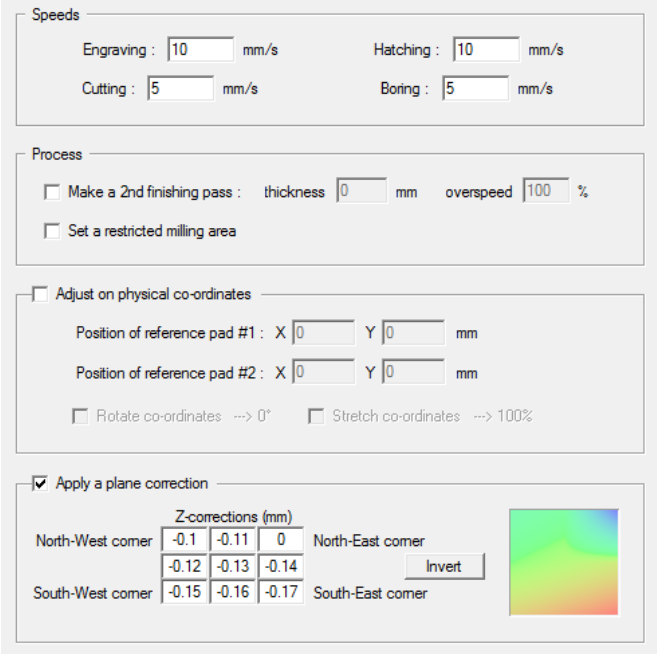


Workpiece origin XY can be located onto a card corner, but keep in mind that Gerber format contains no information about card dimensions and shape. Consequently, it is often better to set the origin on a given pad.

Workpiece origin Z can be located on top or bottom surface of the card. **Is you choose the bottom surface, then do not forget to enter the card thickness** in "File / Dimensions".

For a **double-sided card**, the typical procedure should be as follows:
1- Generate the output file that corresponds to one of the side, including drills.
2- Mill that file, wherever the workpiece origin point has been set.
3- In CopperCAM, locate the origin point on a drilled pad which hole is easy to find on the reversed card.
4- Generate the output file that corresponds to that other side, including cut-outs. You will probably need to mirror X for these sections.
5- Reverse the card, set the physical workpiece origin on the drill you have selected at step #3 which is now the design origin, and start milling.

Milling output window contains **advanced parameters** that you can access using the appropriate button.



Feed speeds that have been set in "Parameters / Selected tools" are displayed for remain and can be overwritten here if necessary.

A second **finishing pass** is possible with a Z thickness margin to be defined. In such case, the milling process will reach the Z target minus that value, and only the finishing pass will reach the normal Z depth. Of course this does not apply to drilling cycles. You can also **define a rectangular milling zone** to redo a limited cycle on a part of the card that could have been machined abnormally at less depth.

If the card layout on the machine bed is not perfectly cartesian, then you may **adjust milling coordinates** on a pair of reference points that have been accurately measured. These points must correspond to CopperCAM reference pads. When milling, that adjustment will rotate or stretch the coordinates so the pads match the measured points.

Last, you may **apply a plane correction** if you have Z measures that correspond to the card corners, midlines or centre on the machine (3 x 3 matrix). These Z points should be taken with a comparator or any measurement device on the machine. CopperCAM does not drive the machine directly and therefore cannot take these points itself. If you have some software or plugin for Z plane correction, either this software will directly modify the output file, or change the driving coordinates. In all cases, this happens beyond CopperCAM. Note that the post-processor format syntax allows you to integrate labels for external variables.

❑ TROUBLESHOOTING

If you have any problems with a reluctant Gerber file, an abnormal output, a possible bug, or want to suggest an improvement, or anything else, the best solution os to contact the developer throughout www.coppercam.com website where a mailing address is displayed. Whatever your request may be, you will get an answer.